

# Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

## Robotics Software Engineer

**Robotics & Software Engineer** with over three years of experience developing real-time perception and control software for edge devices using C++, Python, and ROS2. Skilled in designing, developing, and debugging real-time robotic applications from prototype to production. At Esbaar, I oversaw the field deployment of a computer vision safety system at a British Petroleum oil rig in Oman. Experienced in developing software for NVIDIA Jetson Xavier AGX and Intel VPUs for on-device AI. Recently completed an M.Sc. in Robotics at the University of Bristol, focusing on real-time perception, software integration, and robotic application development and debugging.

## WORK EXPERIENCE

### Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

#### AI Engineer

- Designed and deployed real-time perception software using YOLOv5 and Python for industrial safety systems, improving worker and equipment detection accuracy by 15%.
- Oversaw on-site deployment of the system at a British Petroleum oil rig in the Omani desert, ensuring deterministic performance and reliability under extreme environmental conditions.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Implemented continuous integration and testing workflows for edge inference modules, improving maintainability and deployment speed across hardware targets.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Collaborated with hardware teams and the client to validate software performance in real-world field tests and enhance system robustness.

### LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

#### Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Developed C++17 inference pipelines to maximize performance in resource-constrained environments, integrating CNN-based detection and multi-object tracking (YOLOv5, DeepSORT) for consistent results in dynamic scenes.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing throughput by 233% and enabling real-time detection of safety-critical events.
- Developed synthetic data generation and augmentation tools using OpenCV and NumPy, accelerating model iteration cycles and improving edge-case coverage.
- Implemented rigorous testing and validation frameworks simulating challenging lighting, occlusion, and motion conditions, improving CNN stability and reducing variance by 25%.
- Deployed a unified quantised CNN model for IR and RGB sensor streams using Intel OpenVINO, ensuring deterministic real-time inference on constrained edge devices.
- Collaborated closely with software, hardware, and QA teams to integrate AI modules into production systems, delivering scalable, fault-tolerant solutions for engineering safety applications.

## EDUCATION

### Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

#### Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and RGB data for radiological mapping on an Unmanned Ground Vehicle (UGV).
- Processed and analysed large-scale point cloud datasets using Open3D and RTAB-Map to perform 3D feature extraction, registration, and map optimisation.
- Implemented the SLAM pipeline in C++ using the Point Cloud Library (PCL) for efficient point cloud processing, filtering, and data association.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS 2 before successful real-world deployment.
- Integrated multiple sensing modalities to achieve robust spatial and radiological understanding, enabling accurate obstacle detection in static environments.

### Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

## CERTIFICATIONS

### Deep Learning Specialisation

deeplearning.ai

Jul 2019

## PROJECTS

### ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated differential-drive robot in Gazebo Harmonic and implemented SLAM and navigation in ROS2 Jazzy, with C++17 nodes using slam\_toolbox and Nav2 (Ceres-based optimisation).
- Implemented a C++17 LiDAR denoiser with PCL-based point cloud filtering, improving SLAM stability and map consistency with modular launch files and parameterised YAML configs for flexible experiments.
- Containerised the system with Docker for reproducible builds and CI integration.
- Implemented RViz visualisation and ROS2 introspection (rqt\_graph, topic hz/echo/info, doctor, TF view) to validate pipelines and TF correctness.
- Enforced real-time reliability by tuning SensorDataQoS and verifying consistent frame IDs with a healthy map → odom → base\_link tree.

## SKILLS

- **Core Machine Learning and Optimisation:** Convolutional Neural Networks (YOLOv5/v8, EfficientDet), DeepSORT, OpenVINO, TensorRT, ONNX, Optuna, MultiRocket, Bayesian Optimisation
- **Perception & Computer Vision:** OpenCV, LiDAR SLAM, Open3D, Point Cloud Library, 3D Feature Extraction, Registration, Denoising, NumPy, Scikit-Image, Data Augmentation
- **Cloud, Edge & MLOps:** AWS, Microsoft Azure, MLFlow, Intel VPUs, NVIDIA Jetson, FastAPI, Apache Kafka, DVC (Data Version Control), Docker, Linux, Git
- **Programming & Development:** Python, C++, Bash, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), Gazebo (Harmonic), Nav2, Pandas, PostgreSQL